Lab 0 – Networking VMs

Background / Scenario

In this lab you will first create 2 VMs and connect them together using a virtual network. You will then connect 2 VMs through a third VM that will act as a gateway. In order to attempt this, you must first have completed Lab 1 Part 1 and have two VMs successfully communicating through an internal virtual network.

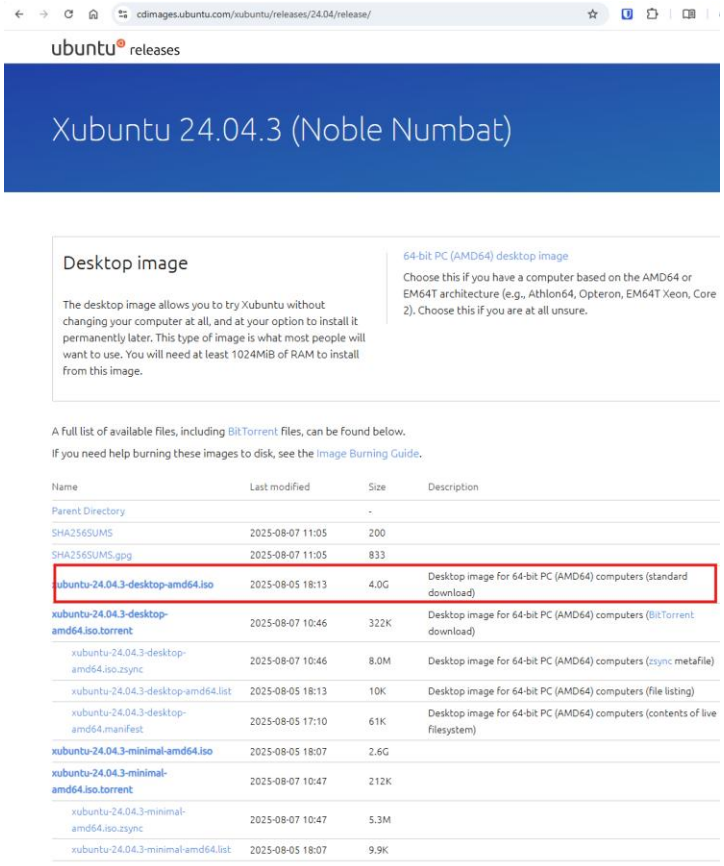
Part 1: Network Two VMs Together

Step 1: Create a VMs and Install a Guest OS

1. Download the Xubuntu ISO for your system from the Xubuntu website.

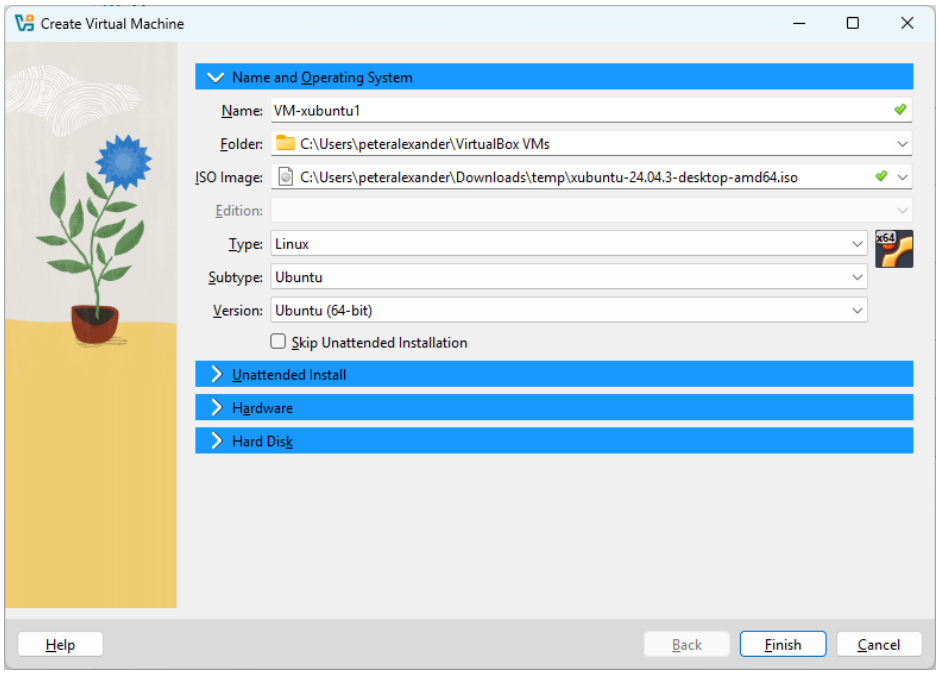
   https://xubuntu.org/download/

   The ISO can be downloaded using Torrent or through a Mirror Server (choose the Mirror download option if completing this using the lab workstations).

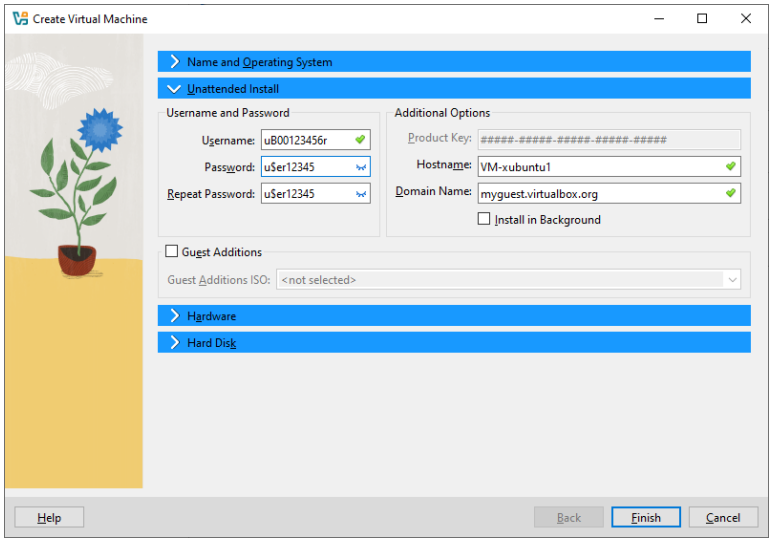


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2. Create a VM in VirtualBox (or an alternative hosted hypervisor such as VMware Workstation/Player) called **VM-xubuntu1** and select your ISO image.
   Note: The instructions here are for VirtualBox but the steps would be similar if using a different hypervisor.



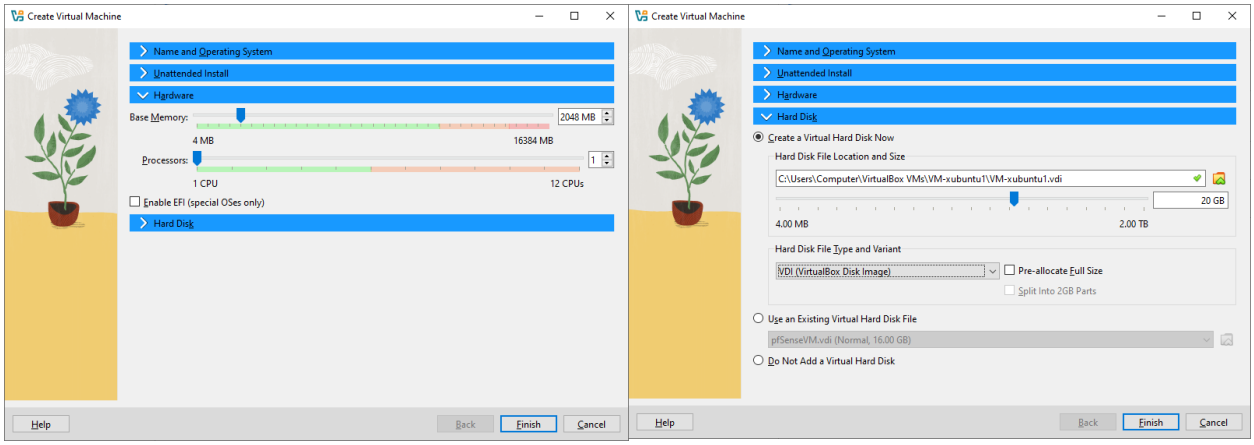
3. Expand the “Unattended Install” section and create a user account with a username of your **u*StudentID*r** (e.g. uB0124564r) and a password of **u$er12345**. (Note: Not a good password to use in a production environment!).



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4. Expand the “Hardware” and “Hard Disk” sections and configure the VM to have 2GB RAM, 1 CPU and a 20GB virtual hard disk.

   Note: These settings are just for the purposes of this lab but could be increased/decreased as needed. See here for exact system requirements.



5. Click **Finish**. Now that the Create VM Wizard is complete, the VM should automatically power on after a few seconds and complete the installation.

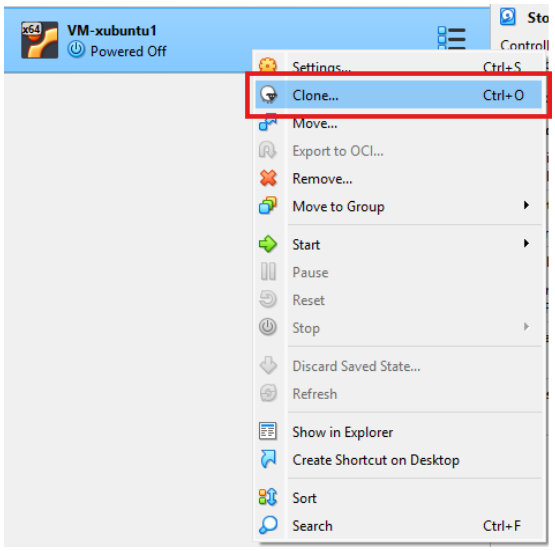


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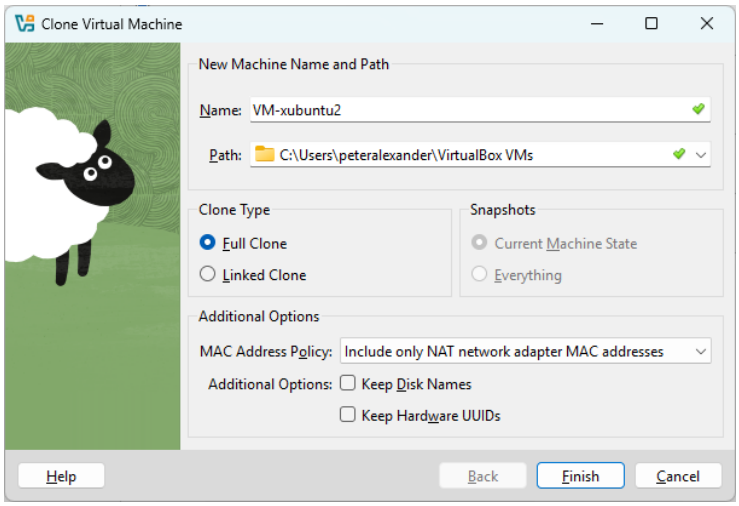
Step 2: Create a copy of your VM

A *clone* of a VM is an exact copy of an existing VM, including the virtual hardware settings, guest operating system and installed applications. Create a second VM called **VM-xubuntu2** by cloning your first VM.

1. Right-click on your VM-xubuntu1 and click on “Clone…”. (You must power off your VM first.)



2. Give the new cloned VM the name **VM-xubuntu2**. Click **Finish**. After a few seconds the new VM should be created.

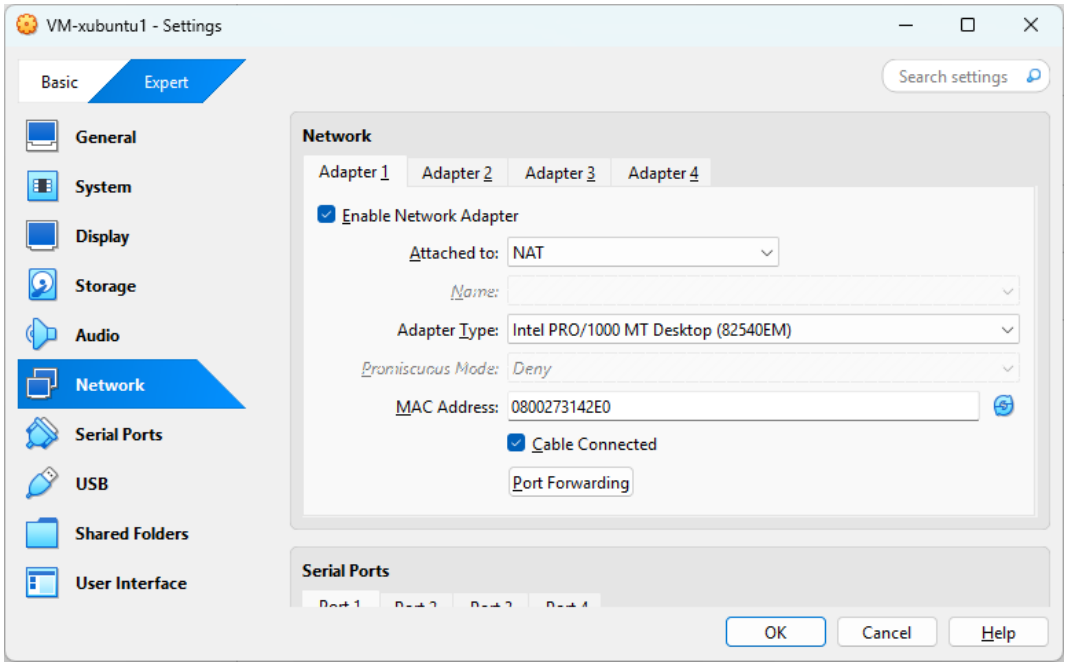


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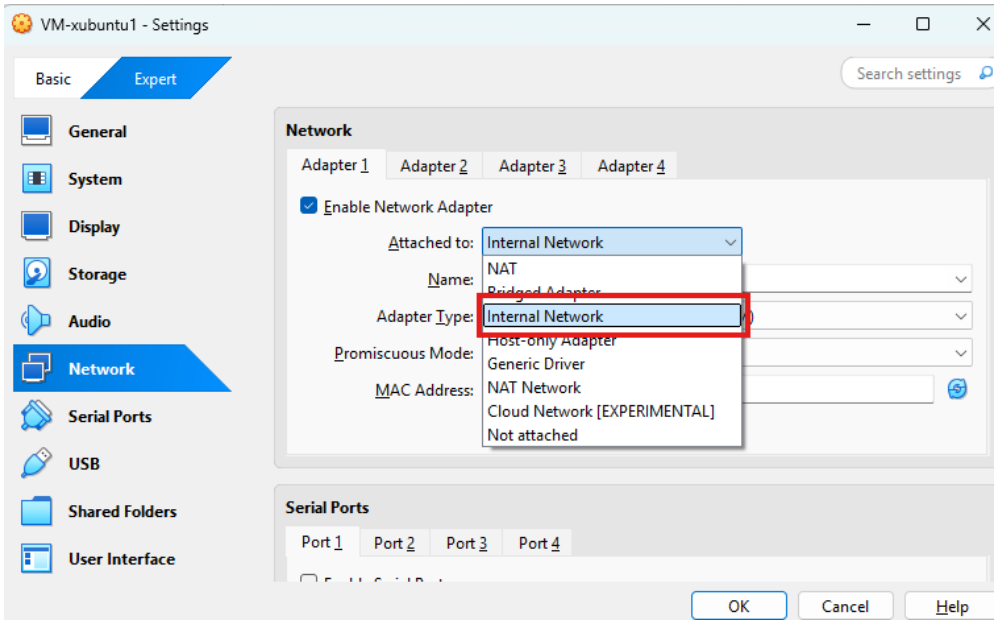
Step 3: Connect your two VMs together

Connect both of your VMs together to the same internal network using the 10.10.10.0 /26 address range.

1. Right click on your first VM and go to **Settings → Network**.

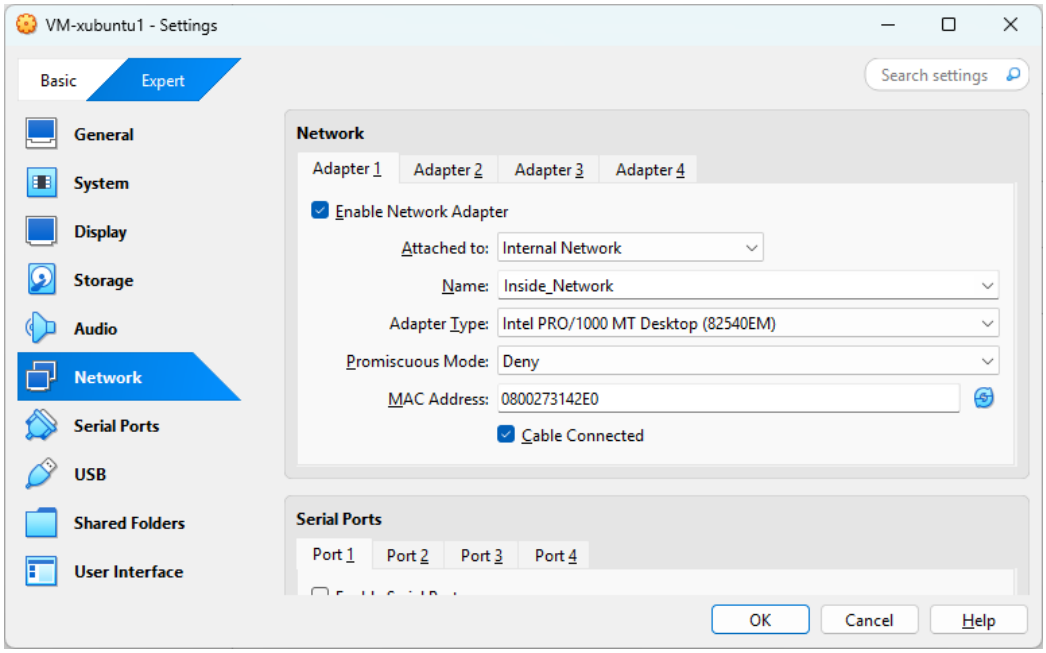


2. Change the network in the “Attached to…” box from NAT to **Internal Network**.



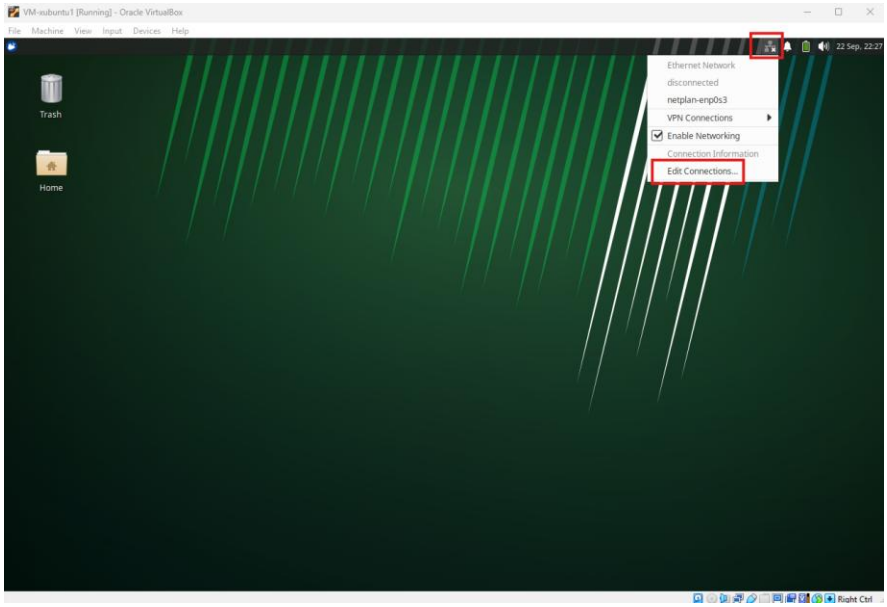
3. Give your internal network a name. It does not matter what you call it, as long as you use the same name for both VMs. In the example below the name “Inside_Network” is used.

   Note: Any VMs attached to the same internal network can communicate with each other (once they have been configured with appropriate IP addresses, etc.). Connecting VMs to the same internal network is equivalent to connecting them to the same LAN switch.



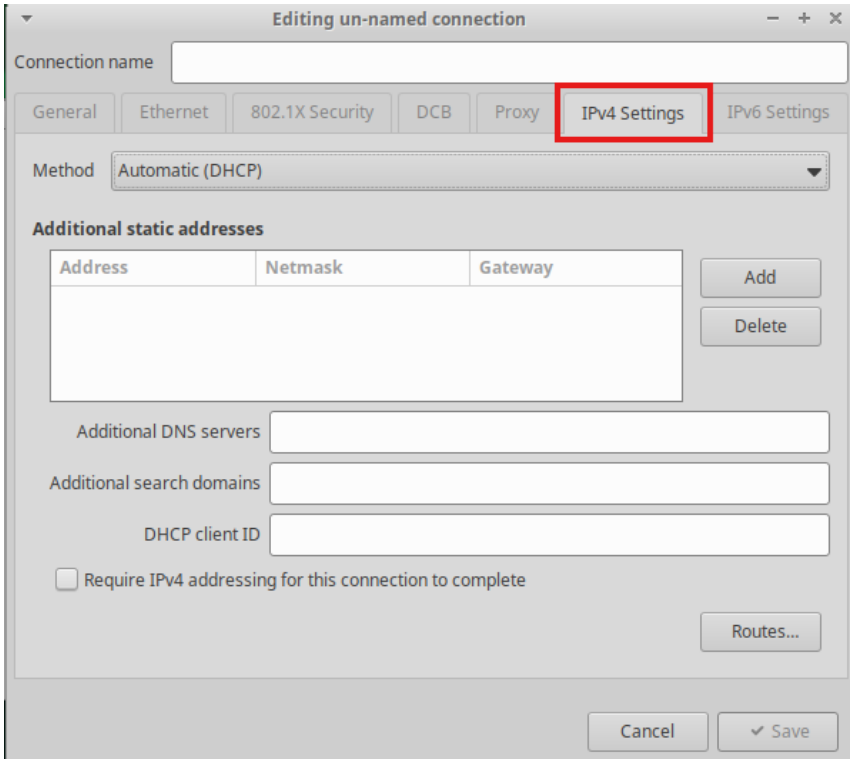
4. Start the first VM and log in using the user details created in Step 1.

5. Go to the network connection settings in your guest OS. There are several ways to do this but one of the most straight forward is to left-click on **Network** in the top-right corner and click on **Edit Connections**.

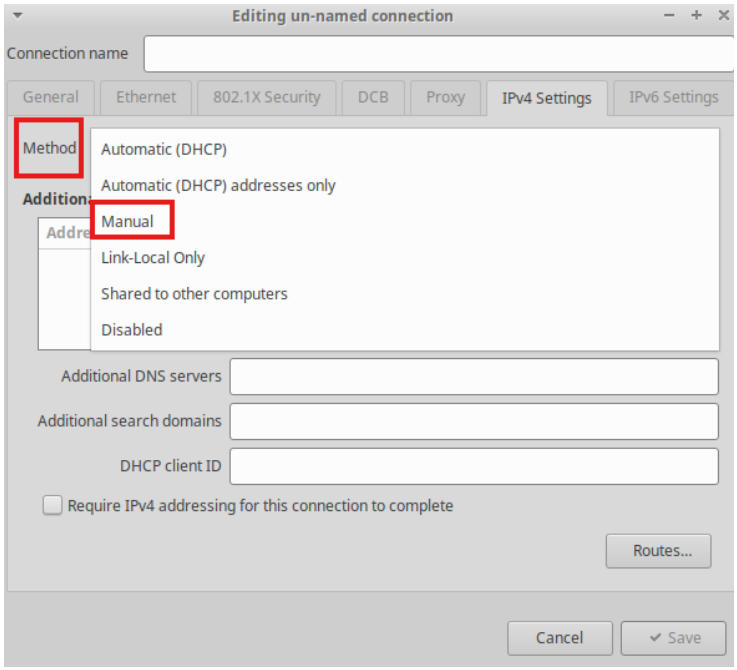


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6. Double click on the network connection. Go to the **IPv4 Settings** tab.

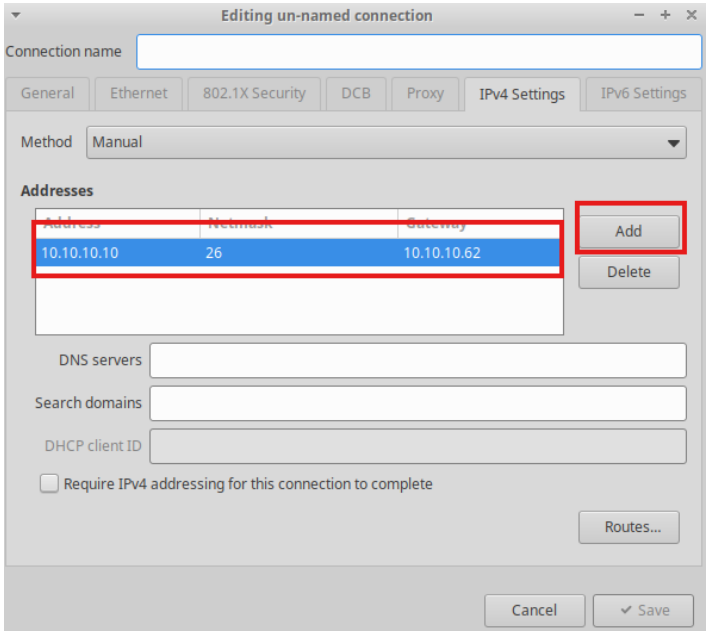


7. Change the “Method” type from DHCP to **Manual**.



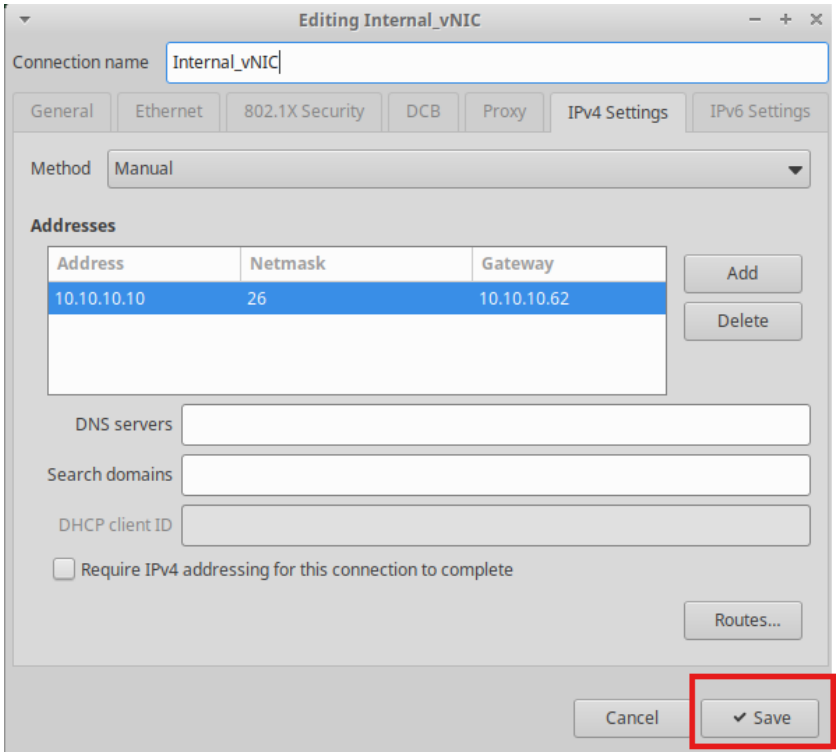
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8. Click **Add** and give the VM an address in the 10.10.10.0 /26 network. You can choose any address but 10.10.10.10 is used below as an example. Use a Gateway address of 10.10.10.62 (there is no actual gateway at this address but you will likely get an error if you leave the gateway address blank).



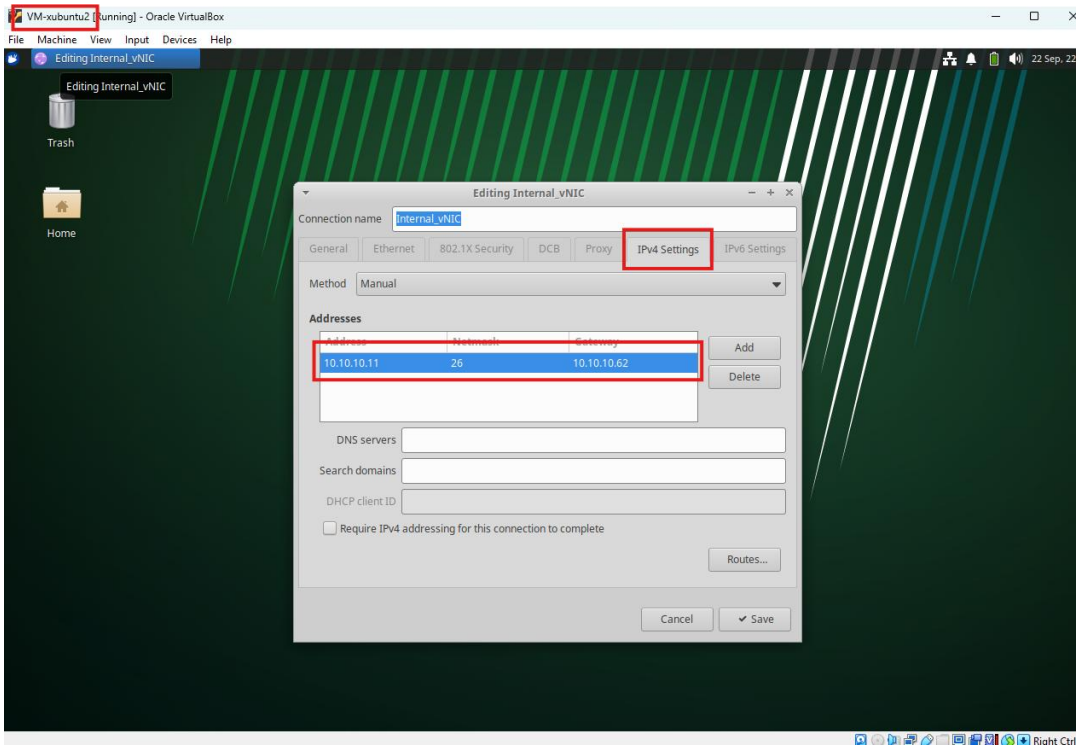
QUESTION 1: What is the dotted decimal equivalent of the /26 subnet mask?

9. Click **Save**. Note: If you are not able to save due to the Connection name being blank, use the name **Internal_vNIC**.

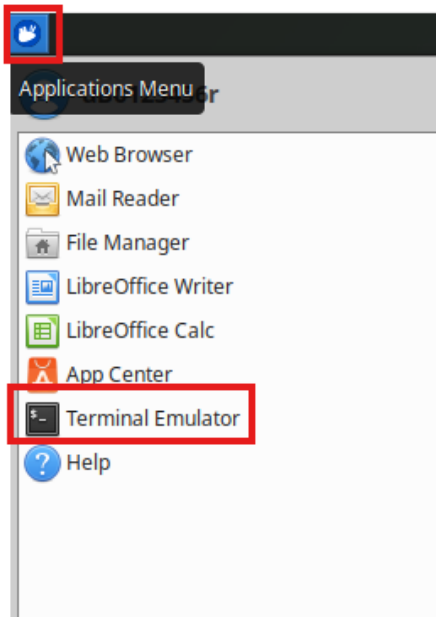


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10. Repeat 1-9 above on VM-xubuntu2, giving it a different IP address in the 10.10.10.0 /26 network (example: 10.10.10.11 /26).

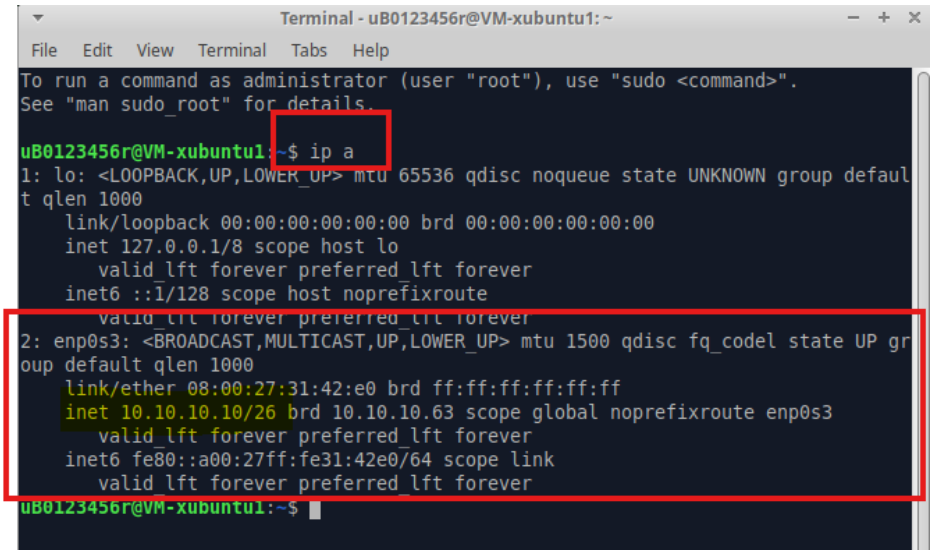


11. Open a terminal window in VM-xubuntu1 by clicking on **Applications Menu** in the top left corner and then **Terminal Emulator**.

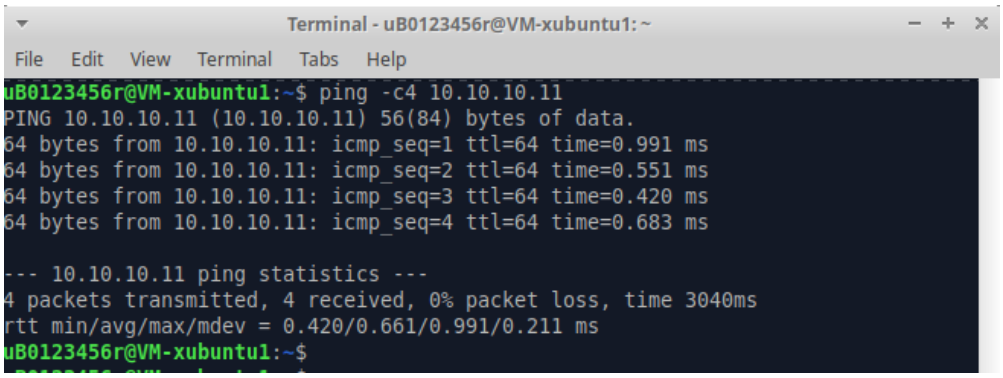


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12. Confirm the IP address of the VM by typing **`ip a`** and pressing enter. Repeat for VM-xubuntu2 also.



13. Attempt to ping from VM-xubuntu1 to VM-xubuntu2 and vice versa.



QUESTION 2: Can your VMs ping each other?

You have completed this part of the lab – please show your lab supervisor the two VMs successfully pinging each other.

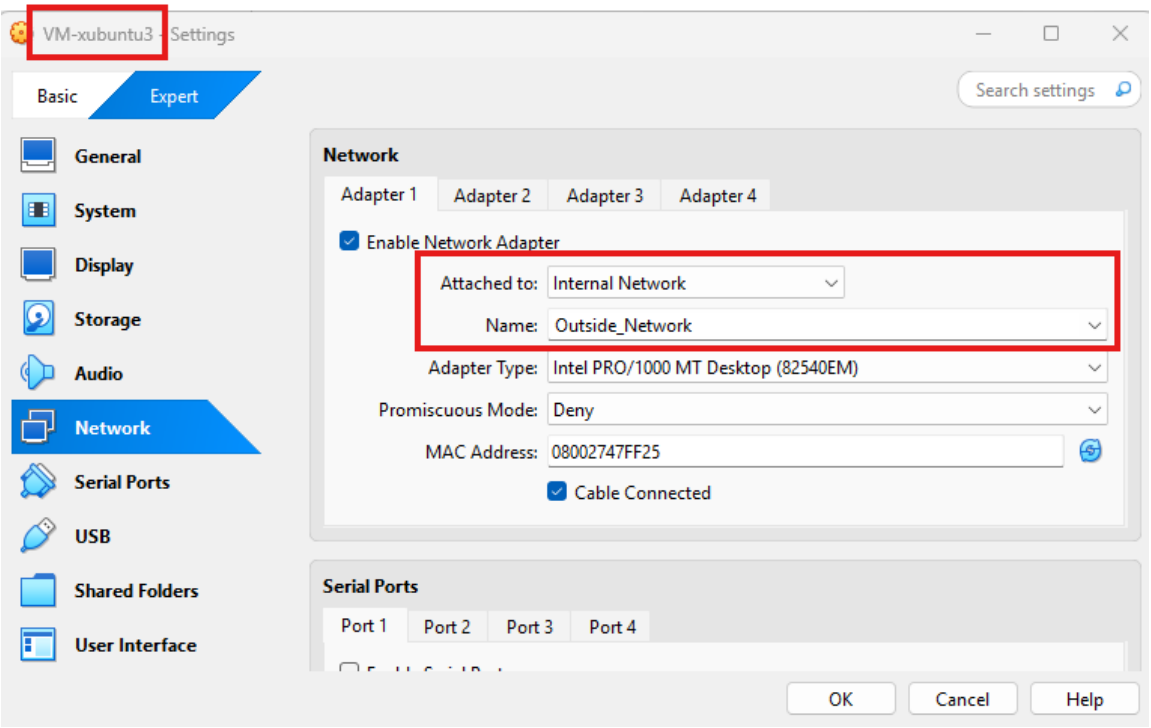
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Part 2: Connect VMs Through a Gateway

Step 1:Create a Third VM and Connect it to a Virtual Network

1. Create a third VM called **VM-xubuntu3** by cloning either **VM-xubuntu1** or **VM-xubuntu2**.
2. Go into the VM settings of your new **VM-xubuntu3** and verify that the network "Attached to" setting is set to Internal Network. Change the name of the internal network to be different to the one that **VM-xubuntu1** and **VM-xubuntu2** are connected together on. In the example below the name "Outside_Network" is used.

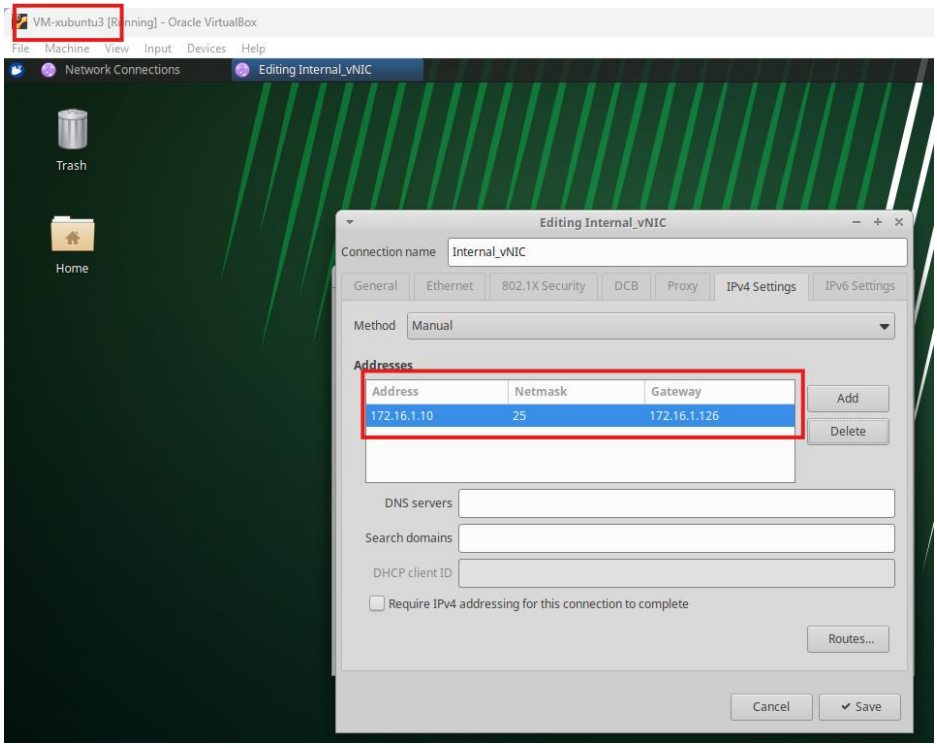
   Note: As outlined in Part 1, if you kept xubuntu3 on the same internal network as the other two VMs, it would be as if all three VMs were connected directly to each other using a (virtual) switch. Since we are ultimately going to force all traffic between xubuntu1 and xubuntu3 to go *through* xubuntu2, we need to connect the VMs to it on two separate virtual networks.



3. Start your third VM and log in (the user details will be the same as the ones configured in Part 1).

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4. Give the VM an address in the 172.16.1.0 /25 network. You can choose any address but 172.16.1.10 is used below as an example. Use a Gateway address of 172.16.1.126.



QUESTION 3: What is the dotted decimal equivalent of the /25 subnet mask?

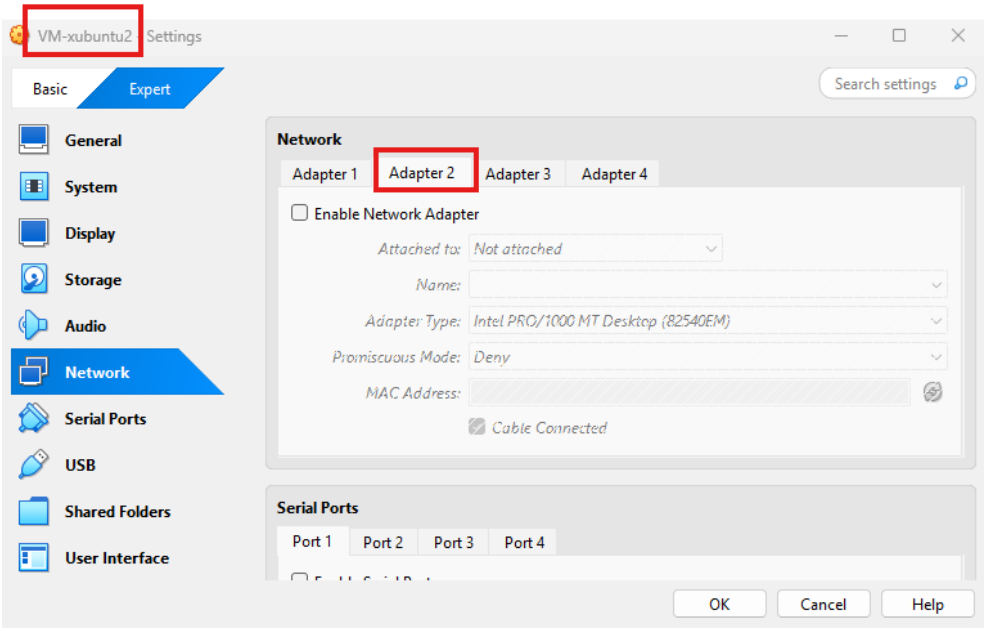
5. Click **Save**. Note: If you are not able to save due to the Connection name being blank, use the name **Internal_vNIC**.

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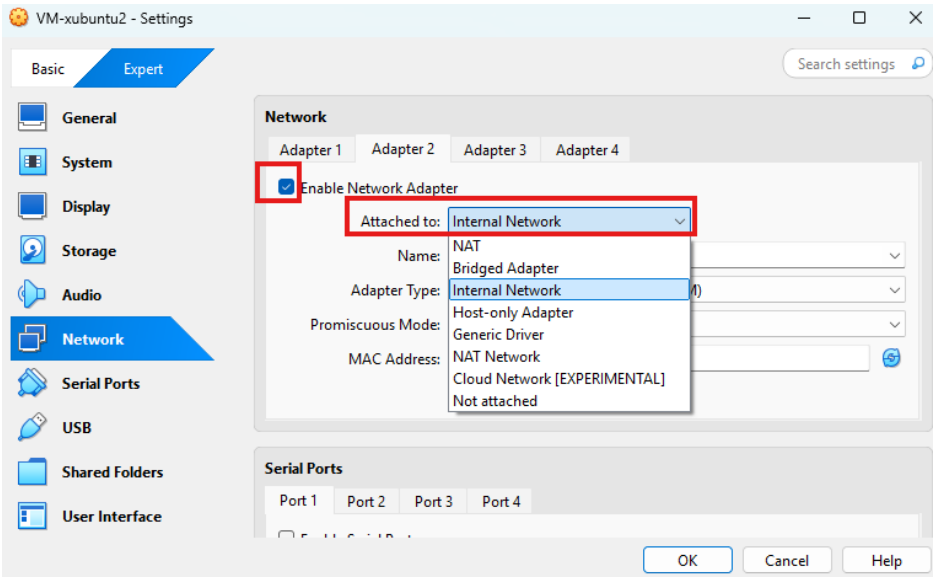
Step 2: Add a 2nd Virtual Adapter to Xubuntu2 & Connect it to Xubuntu3

Most VMs can be configured to have multiple virtual network adapters (vNICs) which allows them to connect to more than one network. The number of vNICs support for a VM is hypervisor dependent but VirtualBox supports up to 4 (and other hosted hypervisors normally support a similar number). In this step you will add a second adapter to xubuntu2 and connect it to xubuntu3.

1. Go into the VM settings for VM-xubuntu2 and go to the Network section.
2. Click on the tab that says Adapter 2. (Note: If it is greyed out, you first need to power off the VM.)

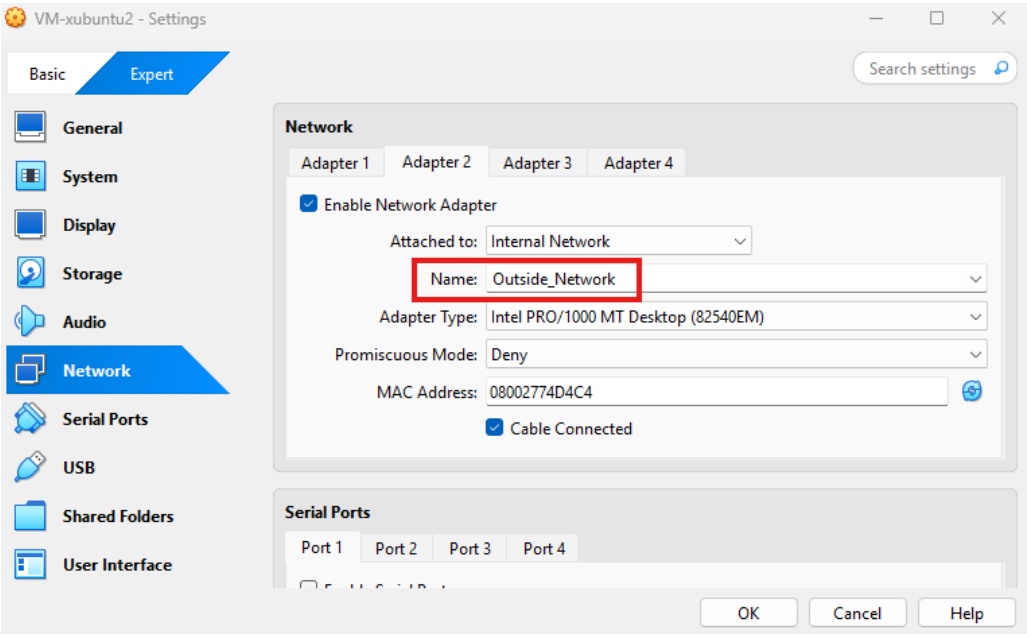


3. Click the **Enable Network Adapter** checkbox and change the “Attached to” setting to **Internal Network**.

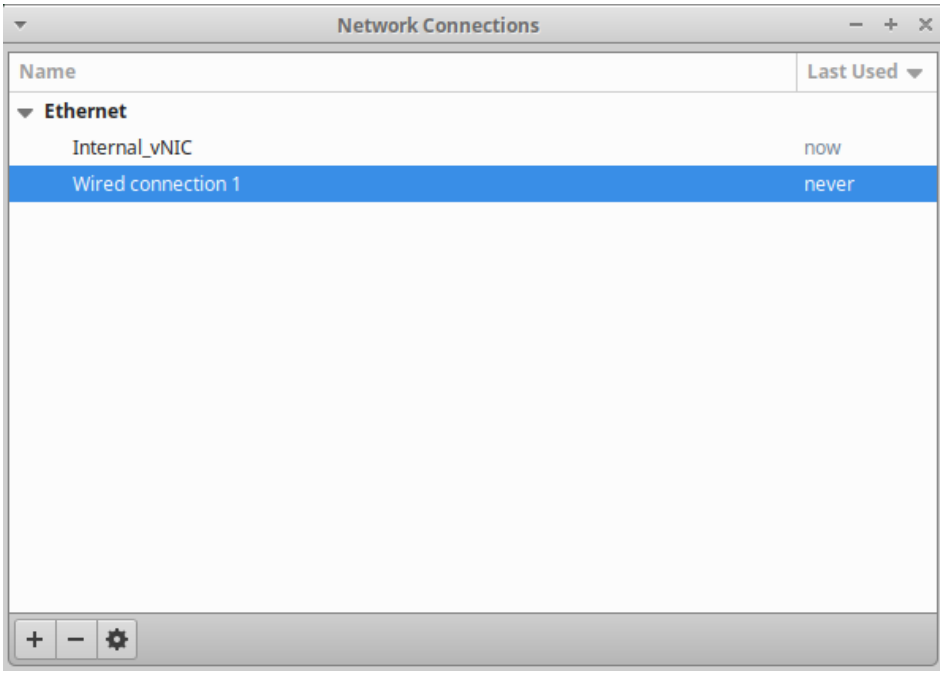


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4. Change the name of the internal network to the same name you attached VM-xubuntu to earlier. In the example below it is Outside_Network. Click OK.

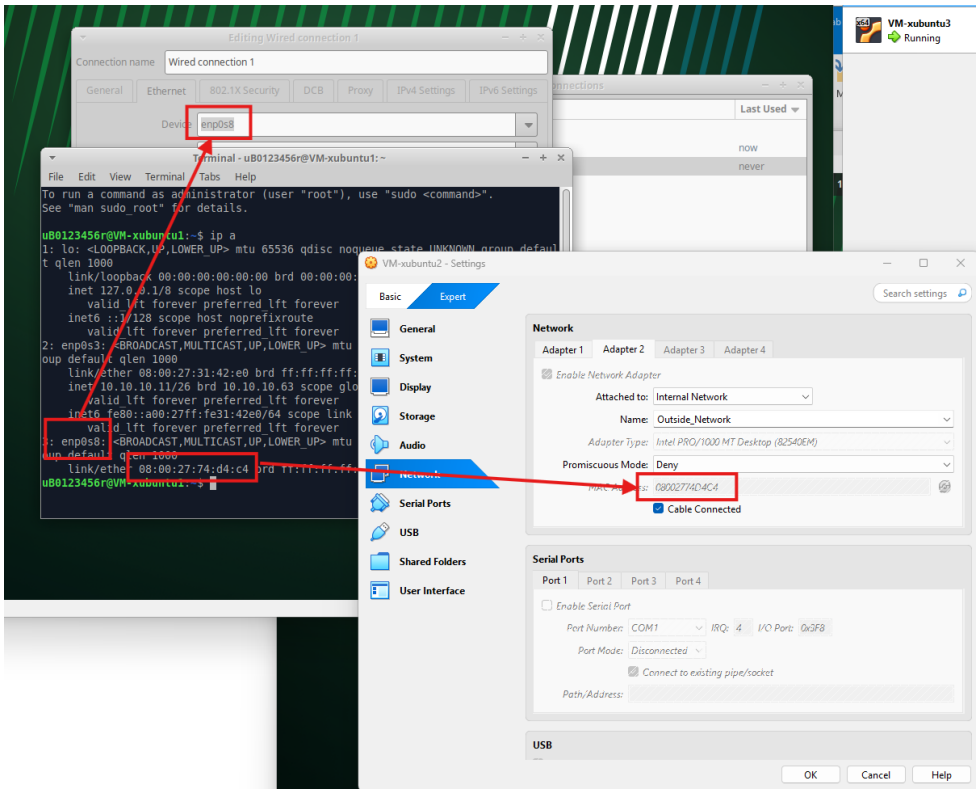


5. Power up the xubuntu2 VM and go to the network setting for your new adapter.

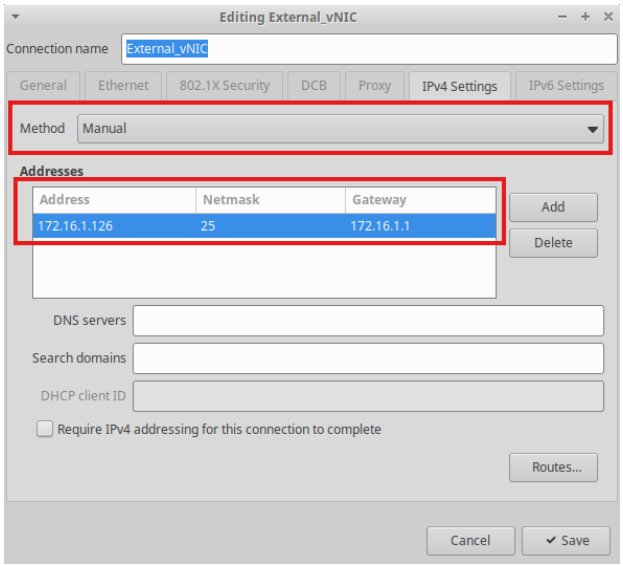


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Note: Now that xubuntu2 has two vNICs, you need to be sure you are configuring the correct one in the guest operating system. If you are not sure, open a terminal and type ***`ip a`*** *to check the MAC address of the adapter and compare it to the MAC address showing in the VM settings for the adapter.*

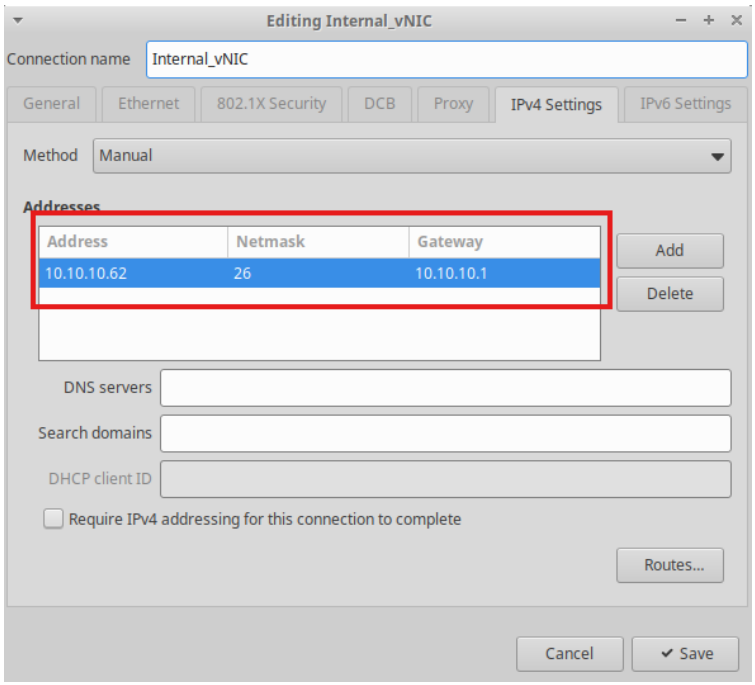


6. Give the second adapter on xubuntu2 a static (i.e. manual) address of 172.16.1.126 /25. (Note: This is the address we specified as the default gateway for xubuntu3.) If you get an error for leaving the Gateway address blank for the adapter, just set it to any unused address in the 172.16.1.0 /25 network. In the example below 172.16.1.1 is used.

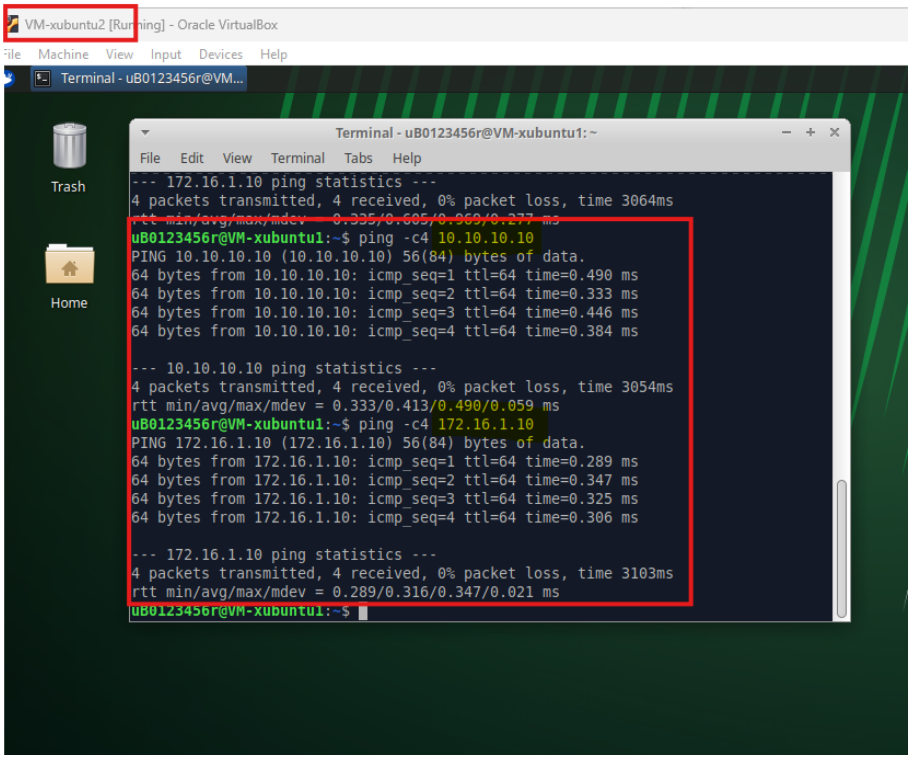


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7. Go into the first adapter on xubuntu2 and change the IP address to 10.10.10.62 /26. (Note: This is the address we specified as the default gateway for xubuntu1 in Part 1.). Change the Gateway address to any unused address in the 10.10.10.0 /26 network. In the example below 10.10.10.1 is used.



8. Open a terminal window in xubuntu2 and verify that you can successfully ping xubuntu1 and xubuntu3. Troubleshoot if either ping is unsuccessful before going any further.



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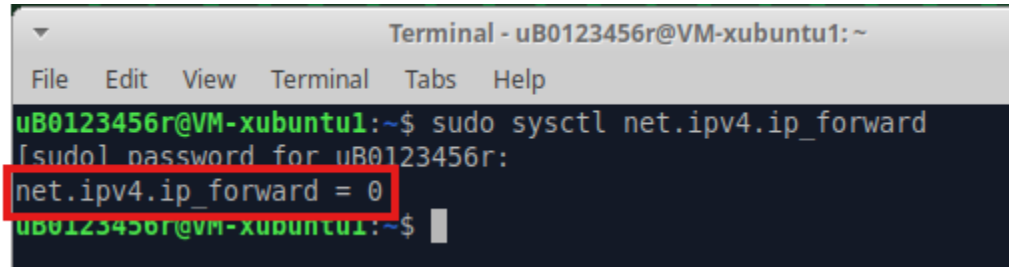
Step 3: Configure Xubuntu2 to be a Gateway Between Two VMs

A Linux based system can be configured to act as a router and forward traffic between different networks. Forwarding is disabled by default on most Linux systems so must be enabled.

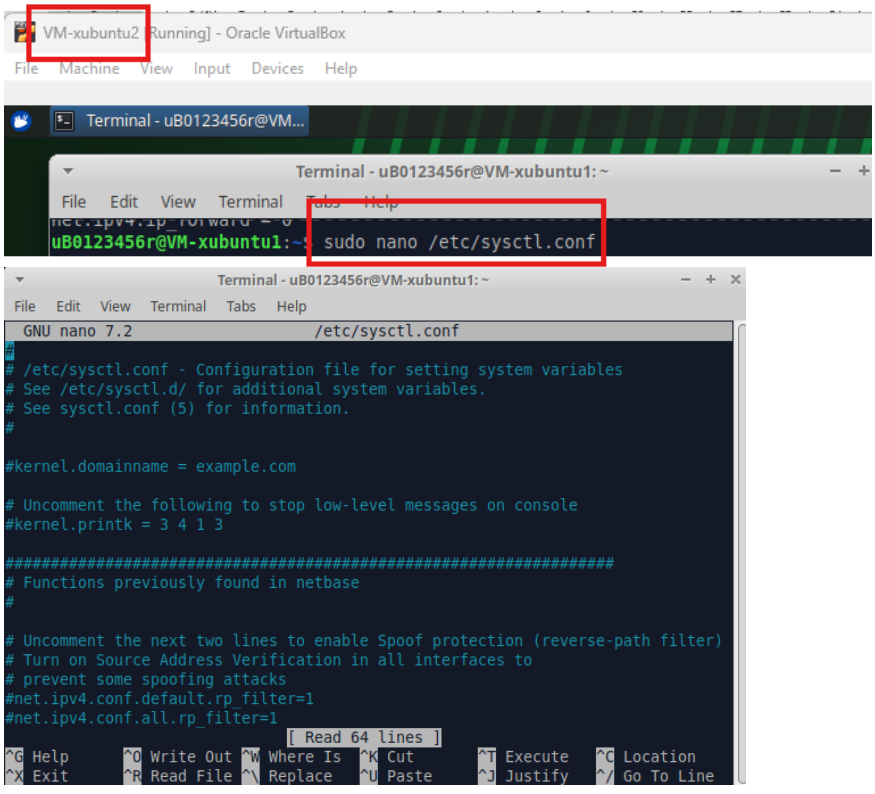
There are a few ways to enable forwarding, for example, it can be enabled temporarily with the command **`echo 1 > /proc/sys/net/ipv4/ip_forward`** entered into the terminal.

In this step, we will permanently enable forwarding on xubuntu2 so that it continues to forward traffic even after a reboot of the OS.

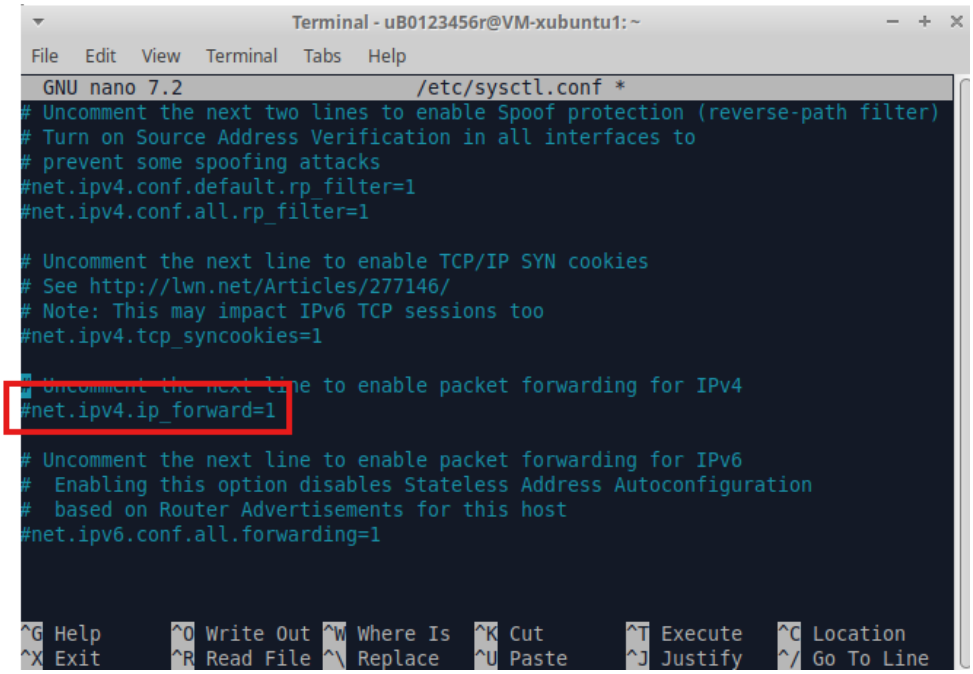
1. Try to ping from xubuntu1 to xubuntu3 (and vice versa). It should fail because, although there is virtual network connections between the two VMs via xubuntu2, the VM has not been configured to forward the traffic between the two networks.
2. Open a terminal window on VM-xubuntu2 and verify that forwarding is currently disabled by entering the command **`sudo sysctl net.ipv4.ip_forward`** into the terminal (the password is the same one you setup in Part 1). A returned value of **0** indicates that forwarding is disabled currently.



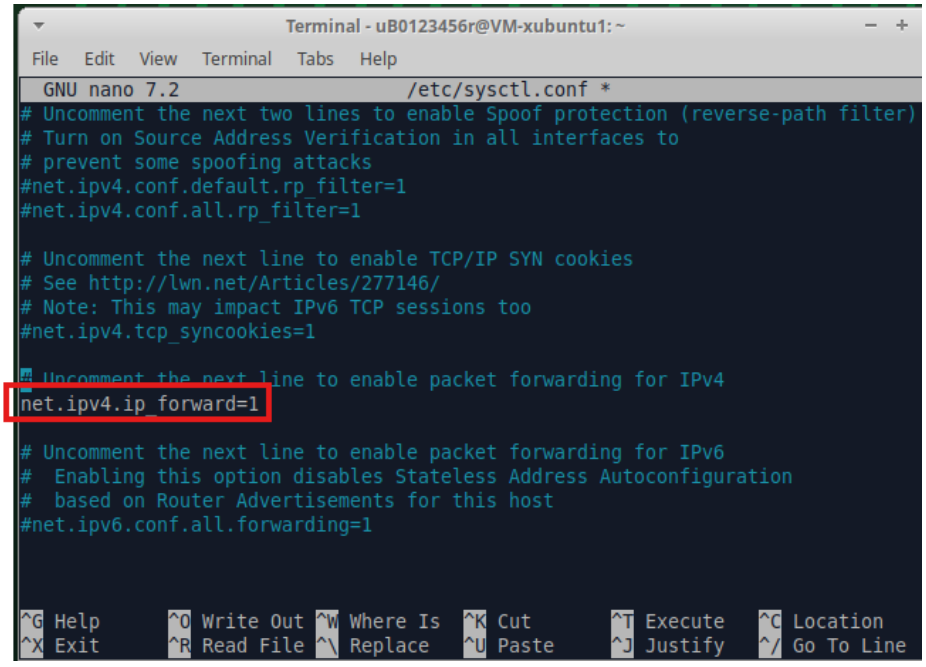
3. Open the **/etc/sysctl.conf** kernel file using the **nano** text editor by entering the command **`sudo nano /etc/sysctl.conf`** into the terminal.



4. Scroll through the configuration file until you find the line relating to **net.ipv4.ip.forward**.



5. Uncomment the line. Save your changes to the file by entering **CTRL+X**. Press **Y** to confirm and press enter to confirm the file name.



*Note: This will allow IPv4 traffic to be forwarded but not IPv6. Notice the separate line in the configuration file for **net.ipv6.conf.all.forwarding** which could also be uncommented to allow IPv6 forwarding.*

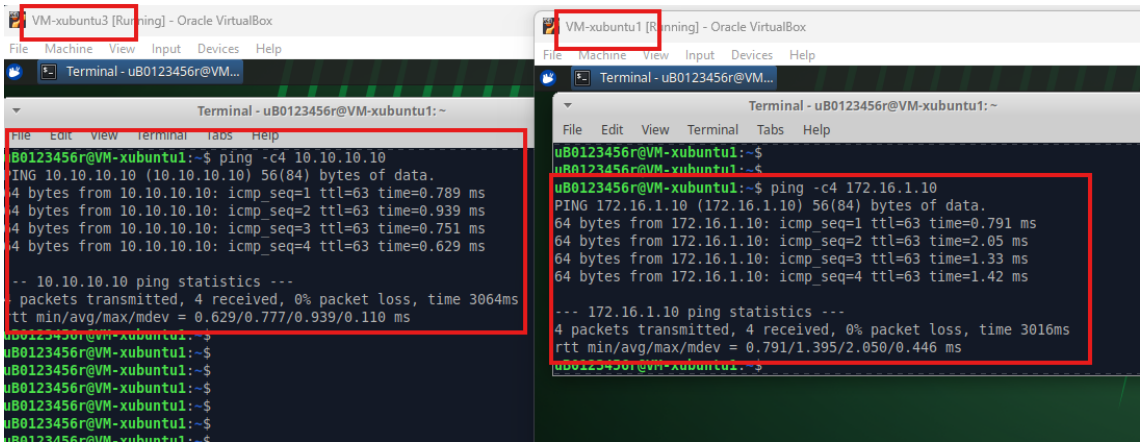
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6. Once the change is saved, enter the command **`sudo sysctl -p`** in the terminal window to apply them. Note: Restarting the VM would also cause the changes to be applied.

```
uB0123456r@VM-xubuntu1:~$ sudo sysctl -p
net.ipv4.ip_forward = 1
```

   A returned value of **1** indicates that forwarding is now enabled.

7. Attempt to ping from xubuntu1 to xubuntu3 and vice versa.



QUESTION 4: Can your VMs ping each other?

You have completed this part of the lab – please show your lab supervisor the two VMs successfully pinging each other.

Additional Challenge: Configure a Connecting from Xububtu2 to the Internet

As an additional challenge, attempt to do the following if you have completed the lab:

1. Add a third vNIC to VM-xubuntu2 and set it as a NAT adapter.
2. Ensure the network connection is set as DHCP.
3. Verify the PC/Laptop you are configuring your VMs on has an active connection to the internet.